

AUTONOMOUS BIOLOGY PLATFORM

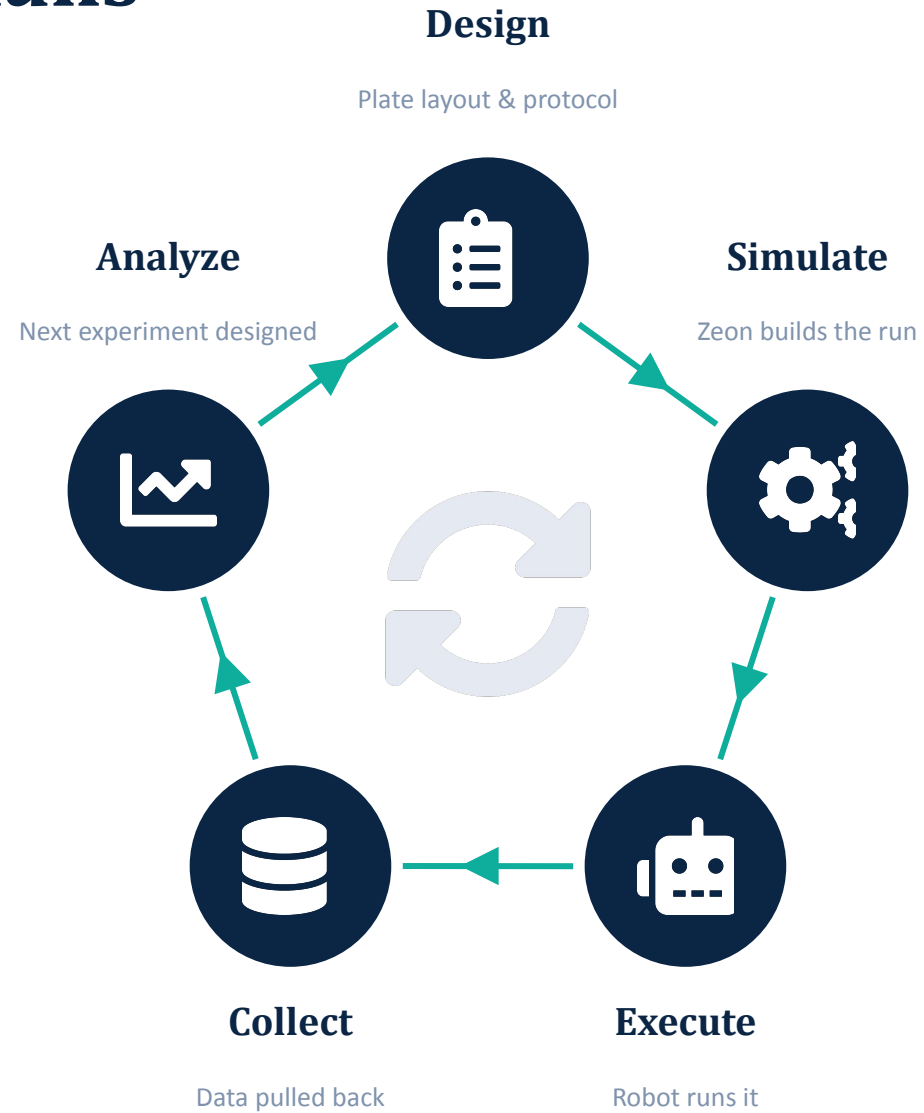
Biomate

Closed-loop automation for biological experiments

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How the Loop Runs

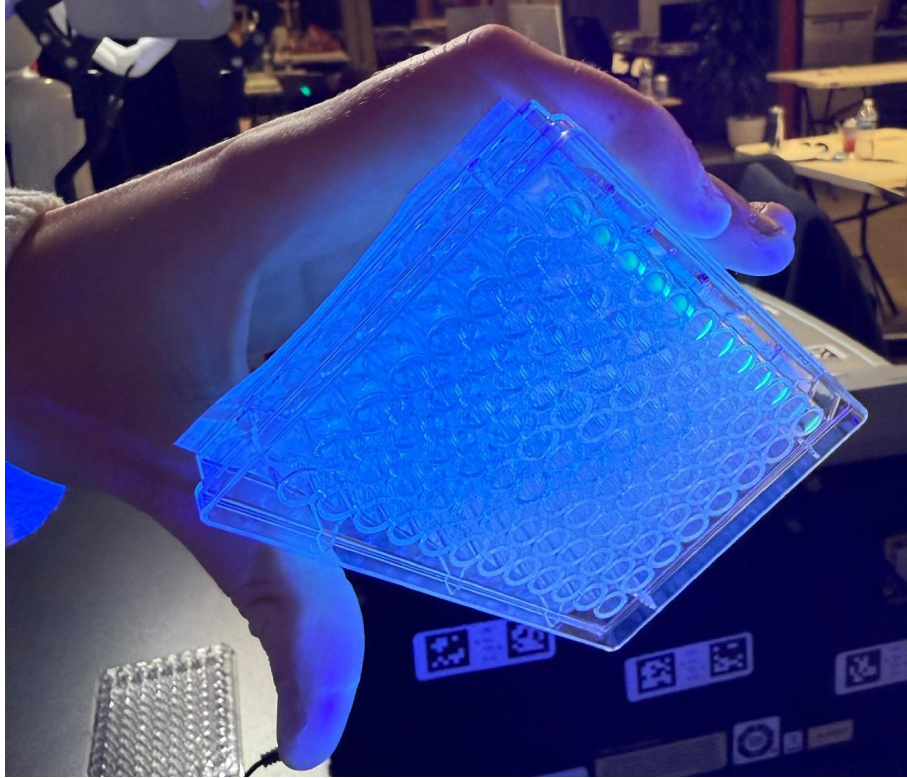


01
EXPERIMENT ONE

Synthesize TEM



- **Design**
Plate layout and protocol for active TEM-1
- **Run**
Zeon simulates it, the robot executes
- **Output**
Purified enzyme, yield and activity data



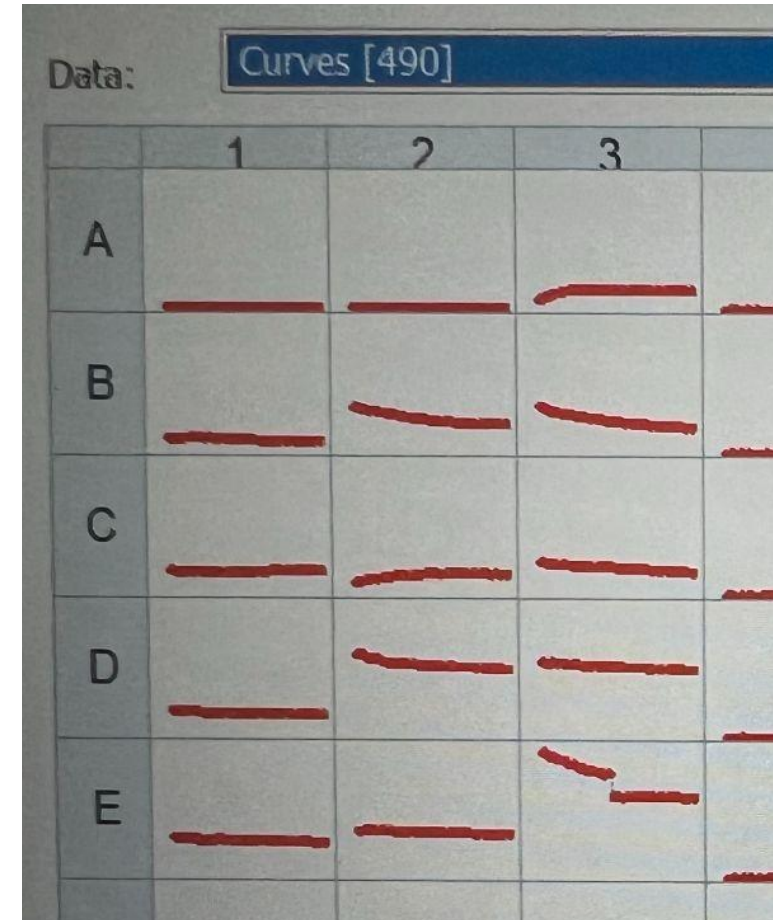
02

EXPERIMENT TWO

Inhibition Assay



- **Setup**
TEM-1 vs. a control β -lactamase
- **Screen**
A panel of inhibitory compounds
- **Readout**
Nitrocefin signal $\rightarrow$ next round designed



Mastermix → Enzyme validation → Agent loop

Edit workflow inputs, deploy to Zeon cloud, open the run URL. After kinetics, the agent escalates dose toward your inhibition target; Z' below threshold re-runs the same experiment.

- 1 · cfps_mastermix
- 2 · cfps_enzyme_validation
- 3 · Agent loop

Stage-1 CFPS expression. Paint the plate (P / N / S), set volumes → Save → Deploy & open Zeon → Confirm & Run.

96-WELL CFPS PLATE

Select a role chip, then Shift+click to paint (or click an empty well). Clicking a filled well only selects it — it will not overwrite Positive/Negative. Maps to pos_wells / neg_wells / sample_wells.

Positive control

Negative control

Sample

Empty

	1	2	3	4	5	6	7	8	9	10	11	12
A	A1 P	A2 N	A3 S	A4 S	A5 S	A6 S	A7 S	A8 S	A9 S	A10 S	A11 S	A12 S
B	B1 S	B2 S	B3 S	B4 S	B5 S	B6 S	B7 S	B8 S	B9 S	B10 S	B11 S	B12 S
C	C1 S	C2 S	C3 S	C4 S	C5 S	C6 S	C7 S	C8 S	C9 S	C10 S	C11 S	C12 S
D	D1 S	D2 S	D3 S	D4 S	D5	D6	D7	D8	D9	D10	D11	D12
E	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12
F	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
G	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
H	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12

1

positive wells

1

negative wells

38

sample wells

MASTER-MIX TUBE PREVIEW

Tube = n × vol_per_well + dead volume. Keep sample tube under ~1500 µL.

30

pos tube µL (1×10+20)

30

neg tube µL (1×10+20)

400

sample tube µL (38×10+20)

BENCH PREP · RECIPE · ROBOT PLAN

Same worksheet as the Zeon cfps_mastermix canvas. Updates live from the plate + volumes above.

VOLUMES & DECK

From workflows/cfps_mastermix.json. Plate wells are set on the left.

Reaction volumes

vol_per_well

10

extra_volume_ul

20

sample_dna_vol

0.2

sample_water_vol

2.8

Cold-block holes

Deck objects

Run options

Save mastermix

Deploy & open Zeon

HANDOFF → ENZYME VALIDATION

pos A1 · neg A2

cfps_mastermix must finish (and plate remain on deck or be remounted) before cfps_enzyme_validation. Controls F4/G4 are taken from the Stage-1 pos/neg wells (A1/A2).

Mastermix → Enzyme validation → Agent loop

Edit workflow inputs, deploy to Zeon cloud, open the run URL. After kinetics, the agent escalates dose toward your inhibition target; Z' below threshold re-runs the same experiment.

1 · cfps_mastermix

2 · cfps_enzyme_validation

3 · Agent loop

96-WELL ASSAY PLATE

Click a well to edit. Shift+click paints the selected role. Saves to data/platemaps/cfps_enzyme_validation_plate.json.

Sample / screen DMSO vehicle Enzyme only DMSO titration Positive control Negative control

Buffer only Empty

	1	2	3	4	5	6	7	8	9	10	11	12
A	A1 S	A2 S	A3 V	A4 E	A5 T	A6	A7	A8	A9	A10	A11	A12
B	B1 S	B2 S	B3 V	B4 E	B5 T	B6	B7	B8	B9	B10	B11	B12
C	C1 S	C2 S	C3 V	C4 E	C5 T	C6	C7	C8	C9	C10	C11	C12
D	D1 S	D2 S	D3 V	D4 E	D5	D6	D7	D8	D9	D10	D11	D12
E	E1 S	E2 S	E3 V	E4 E	E5	E6	E7	E8	E9	E10	E11	E12
F	F1	F2	F3	F4 N	F5	F6	F7	F8	F9	F10	F11	F12
G	G1	G2	G3	G4 P	G5	G6	G7	G8	G9	G10	G11	G12
H	H1	H2	H3	H4 B	H5	H6	H7	H8	H9	H10	H11	H12

SELECTED WELL

Select a well on the plate.

WORKFLOW INPUTS

From workflows/cfps_enzyme_validation.json.

Assay volumes

enzyme_vol

20

sample_vol

2.25

col12_buffer_vol

2.75

vehicle_dmso_vol

2.5

vehicle_buffer_vol

2.5

enzyme_only_buffer_vol

5

buffer_well_vol

25

substrate_vol

25

control_vol

20

control_buffer_vol

5

Incubation & plate reader

From cfps_mastermix (Stage-1)

Deck objects & cold-block holes

Reset plate

Save

Deploy & open Zeon

Set loop policy **before** screening. After kinetics land: QC gate → per-well %inhibition → IC50 curve → next-run decision. Z' fail → **re-run same experiment**. IC50 fit (≥3 doses) sets the next μM; otherwise escalate/reduce.

LOOP SETUP

Max agent loops

Max inhibition % (target_hi)

Min inhibition % (target_lo)

5

90

85

QC gate · min Z'

Starting concentration (μM)

Kinetic JSON path

0.5

225

/Users/arnabbanik/Code/tem1-

Save loop setup

Saved. max_loops=5, band 85–90%, min Z'=0.5

QC GATE

Z' 0.5521

PASS

threshold ≥ 0.5

Pos 0.00029 ± 0.00004 · n=2 · CV 14.4% · r² 0.91

Neg 0.00000 ± 0.00000 · n=4 · CV 70.7% · r² 0.05

Window |Pos-Neg| 0.000283

high_neg_cv

low_neg_kinetic_r2

91.26%

primary %inhibition

11 screening wells · 3 compound(s) · primary Clavulanic Acid 91.26%

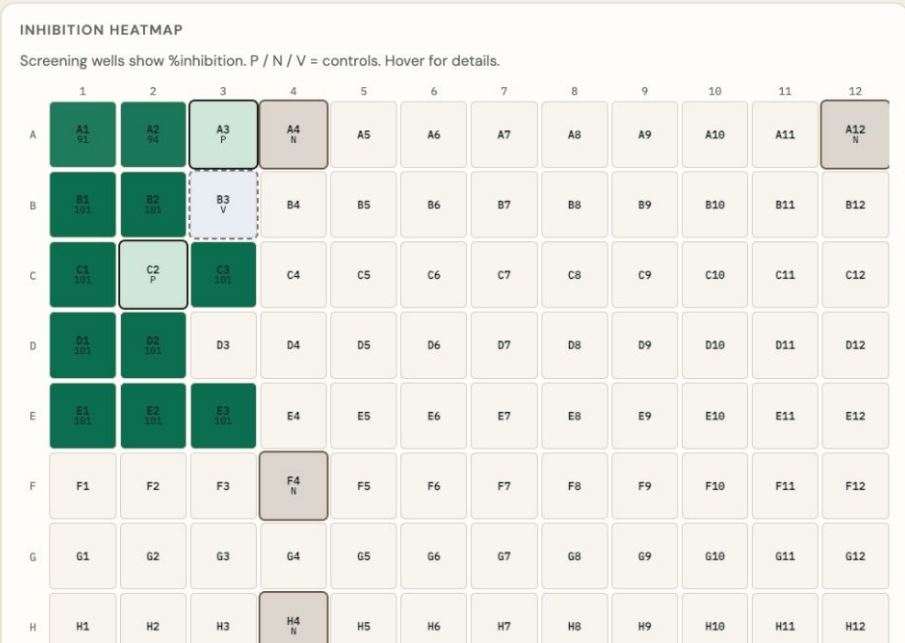
reduce_concentration → 250 μM

Best: Clavulanic Acid (#2, mean 91.26%) · 91.3% above band — halve concentration.

Synthesize + agent step

Reset loop

Re-deploy enzyme WF



COMPOUND RANKING

2 compound(s) ranked. Best: Clavulanic Acid (#2, mean 91.26%). Clavulanic Acid is closest to the 85–90% band (mean 91.26%).

#	COMPOUND	MEAN %	MAX %	N	MM	BEST WELL
1	best Tazobactam	94.1	94.1	1	450.0	A2 (94.1%)
2	best Clavulanic Acid	91.3	91.3	1	450.0	A1 (91.3%)

Well	Role	Compound	μM	Slope	r^2	%inh
B1	sample	Clavulanic Acid	450.0	-0.00000	0.01	100.6
B2	sample	Tazobactam	450.0	-0.00032	0.99	100.6
C1	sample	Clavulanic Acid	150.0	-0.00007	0.93	100.6
C3	sample	Tazobactam	150.0	-0.00014	0.97	100.6
D1	sample	Clavulanic Acid	50.0	-0.00011	0.70	100.6
D2	sample	Tazobactam	50.0	-0.00034	0.92	100.6
E1	sample	Clavulanic Acid	16.7	-0.00007	0.30	100.6
E2	sample	Tazobactam	16.7	-0.00003	0.30	100.6
E3	sample	Enmetazobactam	450.0	-0.00068	0.98	100.6
A2	sample	Tazobactam	450.0	0.00002	0.79	94.1
A1	sample	Clavulanic Acid	450.0	0.00003	0.80	91.3

IC50 · DOSE-RESPONSE



Insufficient points for IC50 fit (need ≥ 3 distinct μM ; have 1). Raise `n_dilution_points` or run more loops. Showing scatter only.

DECISION TIMELINE

- `reduce_concentration`

91.26% @ 450 μM → next 225 μM

Clavulanic Acid is closest to the 85–90% band (mean 91.26%). 91.3% > 90% — halve final μM .

DATA CHAT

Ask about this plate's QC, ranks, %inhibition, IC50, or next run. Grounded in live analysis.

- Best compounds
- QC status
- Next run
- Top compound wells

gemini-2.5-flash

The next experiment should **reduce the concentration** of Clavulanic Acid. The suggested final concentration is **250.0 μM** .

Which compounds rank highest on this TEAM2 plate and why?

gemini-2.5-flash

The QC for this plate **passed** with a Z' of 0.5521.

The highest-ranking compounds are:

- * **Tazobactam** (T1262) ranks #1 with a mean of 94.1% inhibition.
- * **Clavulanic Acid** (T19860) ranks #2 with a mean of 91.26% inhibition.

They rank highest because they show the highest mean %inhibition among the tested compounds.

Ask about this experiment...

Send